

Instructor: Date:

Learner: Regional Branch:

Program: Center:

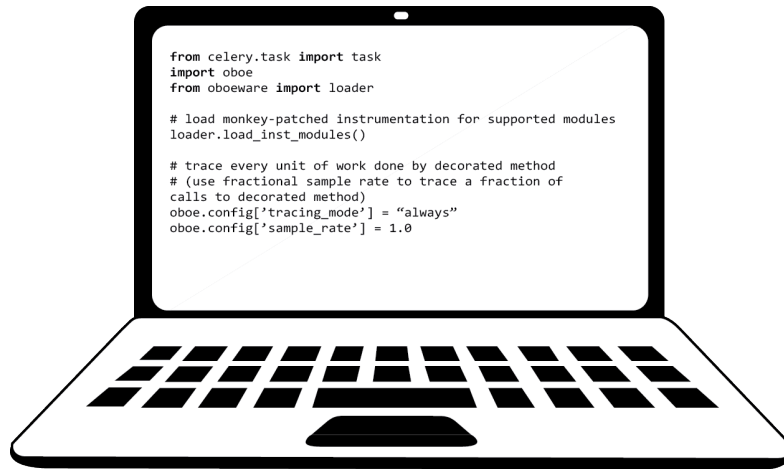
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Let's Start!

A Programming Language with Many Uses!



Learning Outcome:

Describe a general-purpose computer programming language, regarding its history and architecture in spoken and written situations.

Ask your partner:

- What programming languages do you know?
- Which do you think is the most useful language programming when designing an application?

Can Do Statements:

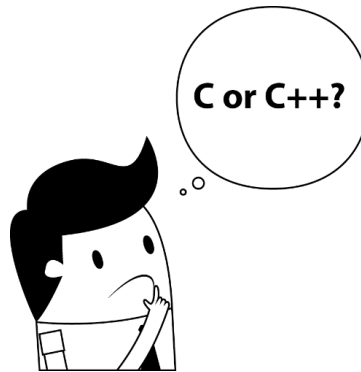
- I can discuss characteristics for C programming language.
- I can exchange information about C++ programming language with a colleague.
- I can present information about other programs based on the C programming language.





Let's Explore!

Paul needs some help for his programming homework.



Instructions: Read the text below.

- Roger:** Hey, Paul. How are you doing? Are you OK? You look worried.
- Paul:** Well, just a little bit. I've got to design my own application using C or C++ programming languages, and I don't even know where to start.
- Roger:** Well, when is it due?
- Paul:** We've got to present an update next Monday.
- Roger:** OK, no problem; let's start with the basics. First of all, C is a general-purpose programming language with object-oriented features. It supports structured programming, lexical variable scope, and recursion. I think you should work on some sort of app for machine instructions. This programming language is perfect for that.
- Paul:** Yes. That's exactly what I'm doing. I think I'll work on that all weekend long.
- Roger:** You're going to have to. Anyway, I recommend you work mostly on C and not so much on C++. C has low-level functions, ideal for mobile apps. Many other languages only interact with C, and I think your code will consist of little more than the interface.
- Paul:** But, why do you think C++ is not a good idea?
- Roger:** Well. In general, C applications are faster to compile and execute than C++ applications. Besides, since its object-oriented programming is not built-in, it is going to give you more freedom to set up structures to act like objects. In addition, in C++ size and memory layout of objects is deterministic.
- Paul:** What about other languages?
- Roger:** C and C++ are way faster than other languages like Python. Another language is Perl. Things are simpler with it as it has only three data types: scalars, arrays, and hashes. However, I don't think you have got to use this language, right?
- Paul:** I don't think so. Anyway, I've got to go now! My next class starts in 5 minutes. Thanks a lot.
- Roger:** You're welcome.



Let's Practice!

Is it a good Idea to work on C?

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. I'm presenting some progress of my project next Monday.
2. I'm not sure, but I think your code will fail eventually.
3. I'm meeting some experts on C and C++ tomorrow morning.
4. I think I'll work on that project all weekend long.
5. I recommend you program in C as it will have low-level functions.

Arrangements at a Specific Time in the Future	Rapid Decisions and Predictions



Let's Practice!

Would you give me some information about C++?

Instructions: Select the correct answer in order to complete the sentence.

1. You are _____ to work on the code all weekend long in order to present some progress.
 - a. have got to
 - b. going to have
 - c. going have
 - d. having
2. I _____ make some progress soon, but I can't decide between C and C++.
 - a. going to
 - b. had going to
 - c. have got to
 - d. got to
3. You _____ to have to think whether working with C++ or not. The size and memory layout of objects is deterministic.
 - a. are going
 - b. got to
 - c. have got to
 - d. 's got



Extension Activity

Let's write about advantages and disadvantages.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about C and C++ programming languages.
- Identify advantages and disadvantages of each one.
- Make a mind map explaining each programming language.
- Share your mind map in the class and discuss about it.

Self-Assessment

Yes

No

- I search for information about C and C++ programming languages.
- I identify advantages and disadvantages of each one.
- I make a mind map explaining each programming language.
- I share information about C and C++ programming languages and discuss about the topic.

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Target Words

- machine instructions** (noun): many bytes in memory that indicate the processor to perform a particular machine operation
e.g. I think you should work on some sort of app for machine instructions.
- recursion** (noun): method where the resolution to a problem is determined by solutions to smaller samples of the same problem
e.g. It supports structured programming, lexical variable scope, and recursion.
- scalar** (noun): atomic amount that is able to contain only one value at a time
e.g. Things are simpler with it as it has only three data types: scalars, arrays and hashes.

- built-in** (adjective): incorporated as an integral part
e.g. Besides, since its object-oriented programming is not built-in, it is going to give you more freedom to setup structures to act like objects.
- general-purpose** (adjective): useful in many ways; not limited to one use or function
e.g. First of all, C is a general-purpose programming language with object-oriented features.
- lexical scope** (noun): part of source code in which a binding of a name with an entity applies
e.g. It supports structured programming, lexical variable scope and recursion.



Learning Points

The Future With Will, Going to, the Present Continuous and the Present Simple

When we want to talk about the future, there are many options; we can use **will**, **going to**, **present continuous** and **present simple**, each one in a particular context or situation.

The **present simple** is used to talk about future events that are clear because they're fixed on a schedule or timetable:

Subject + Verb (-s or -es with he/she/it) + Complement

Example:

*The programming course **starts** next Tuesday at 10:00 a.m.*

The **present continuous** is used to talk about arrangements for events (plans) at a time later than now. It is normally used when there is a very high degree of probability due to a near future situation:

Subject + am/is/are + Verb (ing) + Complement

Example:

*She **is finishing** the program set up tomorrow night.*

Will is used in different situations; when we refer to rapid decisions, offers, promises or quick thoughts at the moment of speaking.

Subject + Will + Verb + Complement

Example:

*I**ll** start the computer while you finish reading that.*

Will is also used to make predictions without evidence:

Example: *Finishing this programming **will** be difficult.*

Be going to is used to talk about plans or decisions made before the moment of speaking, or when there is an intention of doing something:

Subject + Going to + Verb + Complement

Example:

*I'm **going to** use the C++ programming language for the development of this app.*

If we want to make predictions based on some kind of evidence, we also use **be going to**:

Example:

She's really good at this. By the end of the day, the program is going to be finished.





Learning Points

Examples

- *Many other languages only interact with C, and I think your code will consist of little more than the interface.*
- *Besides, since its object-oriented programming is not built in, it is going to give you more freedom to set up structures to act like objects.*
- *Well, when is it due?*
- *That's exactly what I'm doing. I think I'll work on that all weekend long.*
- *My next class starts in 5 minutes.*

Have Got to and Going to Have to for What Is Necessary

Have got to and **going to have to** are used to express an obligation that, for external circumstances, is necessary.

These modal verbs work in different tenses: present and future In the present simple tense, use **have got to** for *I, you, we, and they*; and **has got to** for *he, she and it*.

Example:

Karla **has got to** finish that programming before noon.

To form these sentences in the future tense, write **going to have to** after the subject and verb **to be**:

Example:

You're **going to have to** work all day on that website.

Examples

- *I've got to design my own application using C or C++ programming languages, and I don't even know where to start.*
- *We've got to present some advances next Monday.*
- *You're going to have to.*
- *But I don't think you have got to use this language, right?*
- *Anyway, I've got to go now!*